

Business Scenario & Decision Questions

- The business context is a project to enhance the claims auditing process and workflow. We have a goal of being able to rapidly review any claim ID and determine if the claim was processed correctly.
- The primary stakeholders of this project are claims auditors, compliance and policy team, and the finance team.
- The primary decision question of this project is: “Was this claim’s coverage decision consistent with the policy in effect on the date of service?”
 - Supporting questions:
 - i. Which regions have the highest rate of claims that reference policies with missing or incomplete metadata?
 - ii. Are there any patterns across any member’s claim history that suggest a systematic policy-application error?
 - iii. Which claims involve a procedure code or plan where the applicable policy is missing, or outdated?

Structured Data Architecture

- The three datasets in our data structure are a mix of public data adaptation with synthetic generation; each dataset is around 100 – 1,000 rows with the exception of the procedure catalog reference dataset which is around 20 rows.
- Datasets:
 - claims_ledger.csv – transaction/event dataset:
 - PK: claim_id, FK: member_id, diagnosis_code, procedure_code
 - patient_accounts.csv – entity/reference dataset
 - PK: member_id
 - procedure_catalog.csv - entity/reference dataset
 - PK: code
- The entity relationship diagram (ERD) is saved in the GitHub repository under:
 - Dsba6171_healthtech_teamfour/architecture/erd_domain_microcosm.png
 - The ERD shows the main relationships between the three CSVs.
- **Intended Analytical Use:** Be able to review a claim from a point in time, and along with the knowledge corpus and context, provide an exact answer on whether the claim was processed properly.

Knowledge Corpus & Metadata

- The knowledge corpus contains 7 distinct documents bridging medical coverage policies, procedure allowances, statutory requirements, and claims auditing procedures:
 - HT-POL-001_v2026.pdf: 2026 Inpatient & Outpatient Coverage Policy (Hierarchical structure, current)
 - HT-POL-001_v2024.pdf: 2024 Legacy Policy (Superseded state for version diversity)
 - HT-GUIDE-002.pdf: Procedure Allowance & Fee Schedule Matrix (Contains substantial data table)
 - HT-PROC-001.md: High Dollar Claim Audit & Escalation Workflow (Standard operating procedures)
 - HT-REG-001.txt: Federal Payer Compliance & Prompt Pay Standards (Public regulatory guidance)
 - HT-MEMO-001.md: Temporary Emergency Room Copay Waiver Directive (Labeled SYNTHETIC INTERNAL)
 - HT-FAQ-001.txt: Examiner Audit & Coding Desk FAQ (Operational query guidance)
- **Key Structured to Unstructured Linkages:**
 - Claims_ledger.csv (procedure_code) + patient_accounts.csv (plan_code) -> Maps to HT-GUIDE.002.pdf to verify allowed amounts and pre-authorization necessity
 - Claims_ledger.csv (billed_amount > \$25,000) -> Triggers audit escalation workflow in HT-PROC-001.md

Business Risk Matrix

Quality Problem	Affected Layer	AI Impact	Business Consequence	Future Control
Miscoded data entries for the column insurance_type under the table patient_accounts	Structured Data Validation / Governance	The AI may associate a member with the wrong insurance type or applicable coverage rules.	A claim could be incorrectly approved, denied, or flagged for review.	Validate insurance_type against expected plan/coverage values and quarantine invalid records.
Outdated documents referencing diagnosis codes	Knowledge Governance / Retrieval	The AI may retrieve an outdated policy and apply	Claims may be evaluated using rules that were not valid on the	Track document version, effective date, and authority

Quality Problem	Affected Layer	AI Impact	Business Consequence	Future Control
		obsolete coverage or review requirements.	date of service, leading to inaccurate decisions or compliance issues.	status and retrieve only the policy valid for the claim date.
Incorrect column type for numeric columns (e.g. float vs integer)	Data Ingestion / Schema Validation	Data may be parsed incorrectly or fail validation or matching used by the AI system.	Claims may be excluded, mismatched, or produce unreliable audit results.	Define and enforce an expected schema/data contract and quarantine records with invalid field types.